

Is plug-in solar the next air fryer?

A few years ago, air fryers took America by storm. They were affordable, convenient, and you could find them on any shelf at the store. Could plug-in solar become clean energy's version of the air fryer?

The normalized concept of cooking in oil, baking, or accepting the soggianness of the microwave was revolutionized by air fryers. Plug-in solar has done just that: taken the accepted but suboptimal rooftop solar and offered an alternative solution. Like the air fryer, plug-in solar can be accessible and affordable for Americans, all while generating clean energy.

Plug-in (balcony) solar contains just one or two solar panels. The system can be plugged directly into a wall socket and immediately supply power to your home. Instead of installing them on a roof, they can be set up on a balcony, deck, fence, or driveway. The energy generated is usually enough to power appliances, around 1200 watts, and if excess energy is generated, it can be fed back into the grid.

Compared to rooftop solar, plug-in solar is a significantly cheaper alternative. [70%](#) of Americans cannot afford rooftop solar, and those who don't own single-family homes are likely unable to install it. Plug-in costs a [fraction](#) of rooftop, saving [hundreds of dollars](#) per year. Plus, affordability is only expected to increase. More legislation will foster market competition, and lower commercial prices. Within the next few years, Americans will be saving [more money](#) on a faster timeline.

Known as balcony solar across the Atlantic, Europeans have been ahead of this technology, like most other clean energy innovations. Germany has been the pioneer, creating [legislation](#) to speed up adoption of plug-in solar for its residents. In the United States, Utah was the first state to pass [legislation](#), distinguishing plug-in solar from rooftop, thereby exempting it from interconnection agreements, laws that require residents to run any solar installments by their utility. Now [eight states](#) have passed laws with dozens of others considering legislation.

As legislation continues to pass, more focus is put on the amount of energy we need as a nation. The grid [cannot handle](#) the sheer amount of energy needed in coming years. While holdups in the federal government prevent clean energy infrastructure, plug-in solar is not reliant on federal policy. State legislatures get to decide the rules, bypassing long policy and permitting barriers. Exemplified by Utah, states can create real change in their state through [bipartisanship](#) and shared interests. Although large renewable infrastructure is necessary, supplements like plug-in solar also contribute to the cause, putting power in the hands of Americans, not just large utility companies or clean tech investors. Using Utah's [H.B.340](#) as a blueprint, states can legalize

excess energy being fed back into the grid. So not only are Americans generating their own electricity and reducing reliance on the grid, they are even supplying the grid with energy.

Once safety standards are adopted and legislation is passed in more states, plug-in solar will hit commercial stores. By being sold in stores, solar panels will be transformed from an out-of-reach, expensive technology, into another appliance that any American can buy. This accessibility is key; in order to advance a clean energy transition, making everyone involved facilitates education and the feeling of being part of a greater cause. Making clean energy as simple as any other purchase lowers one of the biggest barriers to adoption. Ensuring that everyday Americans, not just energy buffs, understand the need for a modernized, more efficient grid will be essential.

If we follow the rise of the air fryer, it is mostly due to their convenience, straightforward use, and cheap price. Plug-in solar checks all of these boxes, and most importantly, reduces stress on the grid through the generation of clean energy. Plug-in solar's ability to be accessible and cheap appeals to the Americans in the same way that air fryers did, and it is exactly this accessibility that will propel the clean energy transition forward.